

William Jackson

Senior Software Engineer – Java (Spring Boot) AWS/Azure/GCP

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PROFESSIONAL SUMMARY

Senior **Java Backend Engineer** with 9+ years of experience designing resilient API platforms, transaction-heavy services, and cloud-native backend systems for **SaaS, enterprise workflow, logistics, and data-driven product** environments. Strong hands-on depth in **Java 8–21, Spring Boot 2.3–3.2, Spring Cloud, Hibernate**, and distributed service design, with consistent ownership of architecture, debugging, modernization, and delivery quality. Proven record improving throughput, reducing failure rates, and stabilizing critical integrations through careful refactoring, observability, and secure API design. Known for solving complex production issues, leading legacy migration efforts, and implementing backend features that scale cleanly under real business load.

SKILLS

- **Backend: Java 8–21, Spring Boot 2.3–3.2**, Spring MVC, Spring Data JPA, Spring Security, Spring Cloud, Hibernate, JUnit 5, Maven, Gradle
- **API & Integration: REST API, Microservices, OpenAPI/Swagger, JWT, OAuth2, RBAC**, webhook integration, third-party service integration, async processing
- **Data & Messaging: PostgreSQL, MySQL, SQL Server, Redis, MongoDB, DynamoDB**, Kafka, RabbitMQ, AWS SQS/SNS, Elasticsearch, OpenSearch
- **Cloud & DevOps: AWS (EC2, EKS, ECS, Lambda, API Gateway, S3, RDS, CloudWatch, IAM), Docker**, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD
- **Observability / Quality / Security: Prometheus, Grafana, Datadog, OpenTelemetry**, structured logging, contract testing, integration testing, rate limiting, audit logging, secrets management

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viaro Networks Inc. – (*Delaware, United States*)

- Architected and delivered high-volume **Java** backend services using **Java 17-21, Spring Boot 3.x, Spring Security**, and **PostgreSQL**, supporting multi-tenant workflow orchestration and reducing average API response time by **34%** through query tuning, caching strategy updates, and request-path refactoring.
- Designed secure **REST APIs** for account, billing, and permissions domains with **JWT, OAuth2**, and **RBAC**, giving product and frontend teams a cleaner contract surface while reducing authorization-related defects by **28%** across quarterly releases.
- Led decomposition of a tightly coupled legacy service into focused **microservices**, separating user management, notification, and reporting concerns so deployments became safer, rollback scope became smaller, and production blast radius was significantly reduced.
- Resolved a recurring deadlock and connection exhaustion issue in production by tracing slow transaction boundaries, reworking repository patterns, and tightening pool settings, which improved database stability during peak usage windows by **40%**.
- Implemented an event-driven processing flow with **Kafka** and retry-safe consumers to handle asynchronous document generation and delivery, eliminating fragile synchronous bottlenecks that previously caused request timeouts and inconsistent completion states.
- Modernized an older **Spring Boot 2.x** service into **Spring Boot 3.x** and **Java 17**, updating deprecated security configuration, validation behavior, and Jakarta namespace changes while preserving business continuity through phased rollout and parallel verification.
- Introduced **OpenTelemetry, Prometheus**, and **Grafana** instrumentation for latency, error rate, and queue-depth visibility, giving engineers faster root-cause analysis during incidents and materially improving alert quality.
- Built a flexible feature-toggle framework for controlled rollout of new backend capabilities, allowing product stakeholders to validate premium workflow features safely before broad release and reducing emergency rollback risk.

- Fixed a subtle data consistency bug where concurrent status updates could overwrite final workflow states, then added optimistic locking and stronger state-transition validation to prevent duplicate downstream execution.
- Collaborated with platform engineers on **Docker**, **Kubernetes**, and **AWS EKS** deployment patterns, helping standardize health probes, configuration management, and autoscaling behavior for more predictable runtime performance.
- Created reusable exception handling, audit logging, and response-envelope components so cross-service behavior became more consistent, security reviews became easier, and client teams saw fewer ambiguous failure cases.
- Mentored engineers on backend debugging, profiling, and migration planning, especially around high-risk refactors and framework upgrades, which strengthened delivery confidence and reduced regressions in shared modules.

Software Engineer
Avantia, Inc. – (Ohio, United States)

October 2018 to January 2021

- Built core shipment, route, and partner-integration services with **Java 11-16**, **Spring Boot 2.x**, **Hibernate**, and **MySQL**, supporting operational planning workflows and improving data processing throughput by **31%** after targeted indexing and batch optimization.
- Developed partner-facing **REST APIs** and internal service endpoints for order intake, tracking, and status propagation, enabling cleaner system interoperability and reducing manual coordination overhead for operations teams.
- Investigated a persistent issue where delayed partner callbacks produced duplicate shipment events, then introduced idempotency validation and message de-duplication logic that cut duplicate processing incidents by **45%**.
- Implemented async job execution with **RabbitMQ** for label generation, dispatch notifications, and nightly reconciliation, replacing fragile cron-heavy patterns with more observable and fault-tolerant processing behavior.
- Added a configurable exception-handling and retry framework so backend services could respond more gracefully to unstable third-party integrations without losing traceability or blocking primary transaction flows.
- Refactored a legacy reporting module that mixed SQL, domain rules, and controller logic into cleaner service and repository layers, making enhancements faster and significantly reducing regression risk during release cycles.
- Supported migration from older **Java 8** services to **Java 11** and newer **Spring Boot** baselines, resolving library compatibility issues, test instability, and deprecated API usage that surfaced during modernization work.
- Introduced **Redis** caching for high-read operational dashboards and reference data, which reduced repetitive database load and improved response times for planning screens used throughout the workday.
- Fixed a hard-to-reproduce timezone bug affecting milestone timestamps across partner regions by standardizing UTC storage, tightening serialization rules, and adding regression coverage for cross-region scenarios.
- Worked closely with DevOps on **Jenkins**, **Docker**, and AWS-based deployment pipelines, improving build consistency and shortening backend release preparation time by **22%**.
- Implemented audit trails and role-aware access checks for sensitive operational actions, strengthening compliance posture and reducing risk around manual override functionality in shared admin tools.

Software Developer
ArnAmy, Inc. – (Florida, United States)

March 2016 to September 2018

- Delivered backend modules for account workflows, transaction validation, and internal approval processes using **Java 8**, **Spring MVC**, **Spring Data JPA**, and **SQL Server**, supporting business-critical processing with strong emphasis on correctness and traceability.
- Created service-layer validation rules and reusable domain components that reduced duplicated business logic across controller flows and made regulatory change requests easier to implement safely.

- Resolved a severe production issue where partial failures during multi-step transaction updates left inconsistent records, then introduced transactional boundary fixes and compensating logic to restore reliable processing behavior.
- Implemented new notification and approval-routing features that connected backend rules with queue-based processing, helping business teams shorten turnaround time for exception handling and manual review cases.
- Improved performance of several expensive reconciliation queries by analyzing execution plans, rewriting joins, and adding missing indexes, resulting in **27%** faster batch completion on peak processing days.
- Strengthened API and internal admin security by tightening role checks, session handling, and audit capture, which reduced unauthorized action exposure and improved review readiness during internal assessments.
- Helped migrate a monolithic code area toward modular service boundaries, extracting reusable business services and making future backend feature work more maintainable for the wider engineering team.
- Diagnosed memory pressure and unstable runtime behavior caused by oversized in-memory report construction, then redesigned export generation into paged streaming flows that prevented repeated service degradation.
- Built integration tests for validation-heavy service paths and database interactions, increasing release confidence and catching edge-case regressions before they reached user acceptance testing.
- Partnered with QA and product analysts to reproduce intermittent defects using production-like datasets, which improved debugging accuracy and led to quicker closure of long-standing backend issues.

Junior Software Developer

October 2014 to February 2016

Centric Tech Inc. – (New Jersey, United States)

- Supported backend development for internal business applications using **Java 8**, **Spring**, **JSP/Servlet** patterns, and **MySQL**, focusing on stable feature delivery, bug resolution, and maintainable service logic.
- Implemented form-processing, search, and approval workflow enhancements that improved internal user efficiency while giving senior engineers a stronger foundation for later platform expansion.
- Fixed recurring null-handling and validation issues that caused avoidable runtime errors in approval flows, then added guard clauses and clearer service validation to improve request reliability.
- Helped troubleshoot a difficult batch-job failure caused by malformed input records and weak parsing assumptions, then improved validation and error reporting so support teams could isolate bad records quickly.
- Contributed to early API integration work for internal services, building adapters and transformation logic that allowed older modules to exchange data more consistently with newer backend components.
- Refactored duplicated utility and service code into shared helpers, reducing maintenance overhead and making defect fixes easier to apply across related modules without introducing divergence.
- Assisted with release verification, log analysis, and SQL troubleshooting during deployment windows, building strong debugging discipline and helping reduce avoidable post-release issues.
- Participated in a legacy cleanup effort that replaced tightly coupled controller-heavy logic with clearer service-layer separation, creating a better base for later modernization initiatives.
- Worked with senior developers to implement new backend features in incremental steps, balancing delivery speed with safe testing practices and improving confidence in small but frequent production releases.

EDUCATION

Bachelor's degree in Computer Science

2010 to 2014

Harvard University

- Built strong foundations in **algorithms**, **databases**, and **software engineering**, applying coursework through practical team projects and production-style system design.